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To the Editor-in-Chief,
IEEE Transactions on Smart Grid

Dear Editor,

We respectfully submit our manuscript, *"Crypto-Induced Latency Denial of Service in IEC 61850 Substation Communications: Cyber-Physical Impact Analysis, Priority Admission Defense, and Post-Quantum Evaluation,"* for consideration as a Research Paper in the *IEEE Transactions on Smart Grid*.

Problem and motivation. As digital substations adopt IEC 61850 GOOSE over Ethernet process buses secured by IEC 62351-6, the cryptographic authentication intended to prevent message spoofing introduces a new availability hazard. We identify and analyze *Crypto-Induced Latency Denial of Service* (CIL-DoS): an adversary floods the process bus with syntactically valid but cryptographically invalid GOOSE frames, saturating the receiver's verification queue and delaying mission-critical trip commands beyond the strict 3 ms Class P2/P3 safety deadline.

Contributions. The paper (i) links the communication delay to power-system transient stability via a numerically integrated SMIB swing-equation model (critical clearing time 201.6 ms), ANSI 50BF breaker-failure escalation, and transformer I^2t stress; (ii) proposes the Priority Queue Admission Controller (PQAC), whose novel core is GOOSE-semantic RFC 1982 `stNum` plausibility gating and verify-before-commit state handling, with proven stability and delay-isolation bounds; (iii) measures (5 reps) a reduction of mean trip latency from 2,388 ms to 760 μs at 0.00% deadline miss on a service-time-calibrated Cortex-A9 relay emulation, with a component ablation and an explicit first-class treatment of the spoofed-MAC limitation ($\approx$95% trip loss), and validates the cryptographic service times on real embedded silicon (ESP32); and (iv) benchmarks the NIST post-quantum signature standards (ML-DSA, FALCON, SLH-DSA) to assess quantum-safe GOOSE authentication under the 3 ms deadline.

Originality and ethical statement. This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere, in whole or in part. All authors have read and approved the submission and agree to its content. The work involves no human or animal subjects. All testbed artifacts, scripts, and firmware are open-sourced to support reproducibility.

Scope fit. The paper addresses communication, cyber-physical security, and protection-system resilience in digital substations—core topics of the *IEEE Transactions on Smart Grid*.

We thank you and the reviewers for your time and consideration.

Sincerely,
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on behalf of all authors
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